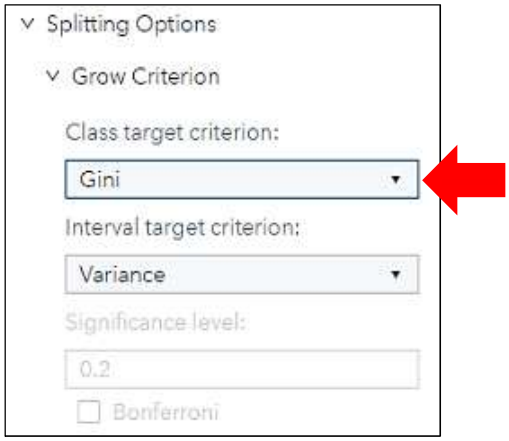
Improving a Decision Tree Model by Changing the Recursive Partitioning Parameters

In this demonstration, you change more settings of the Decision Tree node in the Lesson 3 pipeline. You modify the recursive partitioning parameters and compare this model performance to the models built earlier in the course.

Recall that the previous model, based on changing the tree structure parameters, achieved an average squared error of 0.0663 on the VALIDATE partition. We will try to improve the model's performance by modifying some of the settings of the decision tree model.

1. Under the Grow Criterion properties, change **Class target criterion** from Information gain ratio to **Gini**.



   The Gini index is calculated by subtracting the sum of the squared probabilities of each class from one. It favors larger partitions.

   Information gain multiplies the probability of the class times the log of that class probability. It favors smaller partitions with many distinct values.

   In due course, you experiment with your data and the splitting criterion.

2. Run the **Decision Tree** node.
3. Open the results for the node.
4. Click the **Assessment** tab. (You might need to expand the Fit Statistics table.)

Target ...	Data Role	Partitio...	Formatt...	Sum of ...	Averag...
churn	TRAIN	1	1	39,590	0.0583
churn	VALIDATE	0	0	16,967	0.0630

   The average squared error for the tuned decision tree model is 0.0630 on the VALIDATE partition. This fit statistic, again, is a slight improvement compared to that of the previous model by changing only the recursive partitioning parameters.

5. Close the Results window.

End of Demonstration

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